BP
Si
Di
offset

SS
DS
ES

Question #1: First Question (10 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	A	B	B	B	B	C	C	C	D

10

1. Address bus width of Intel 8086 microprocessor is:
A. 16　　B. 24　　C. 20　　D. 32

2. The physical address of the source operand in the instruction MOV AX, [BX]. Assuming DS register= 2000H, and BX register= 150H, is:
A. 20150H　　B. 2150　　C. 20000　　D. 200150

3. The segment that contains the assembly language instructions to be executed by the microprocessor is
A. Data segment　　B. Code segment　　C. Stack segment　　D. Extra segment

4. Instruction Pointer (IP) contains offset address of ________ segment.
A. Data segment　　B. Code segment　　C. Stack segment　　D. Extra segment

5. If offset is a 13-bit number address, then the maximum possible locations are_____.
A. 4 KB　　B. 8 KB　　C. 64 KB　　D. 16KB

6. The equivalent of instruction XOR BL,FFH is :
A. NEG BX　　B. NOT BL　　C. AND AL,0　　D. OR AL,FF

7. The content of register BL after execution the instruction MOV BL,-25:
A. BL=E5H　　B. BL=F2H　　C. BL=E7H　　D. BL=F5H

8. In the following instruction sequence, show the value of AL, in binary:

```
MOV AL,5
NEG AL
XOR AL,FFH
```

A. AL= 11111011B　　C. AL= 00000100B
B AL= 11110101B　　D. AL= 01011111B

9. Find the content of AL and zero flag (ZF) after execution of the following code:

```
MOV AL,58H
TEST AL, 4
```

A. AL=0 ; ZF=1　　B. AL=58H ; ZF=0　　C. AL=58H; ZF=1　　D. AL=0 ; ZF=0

10. If the SP register contains the hexadecimal value F200H, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
XCHG AH,BH
POP AX
POP BX
```

A. F202H　　B. F208H　　C. F204H　　D. F200H

Question #2 (12 points)

A. ARR DW 3333H, 4444H, 1111H, 2222H, 0FFFH
MOV BX,OFFSET ARR
MOV SI, 6
 	
MOV DI,2
 	
MOV DL, [BX][SI]
MOV AL, [BX][DI]
XCHG AL, DL
 	
AL= D4pC(1) ; DL= 44H
	B. MOV AL, 22H
	AND AL, 03H
	JZ YES
	NOT AL
	JMP EE
	YES: NEG AL
	EE:HLT
	AL= ~~EE~~ FD

C. ARRAY DB 8, 3, 2, 4,6,8,5,7,0
MOV AL,0
MOV CX,4
MOV BX, OFFSET ARRAY
L: STC
ADC AL, [BX]
ADD BX,2
LOOP L
MOV AH,[BX]
 	
AL= 19 H AH= 00 H
	D. X DW 104FH, 201AH, C030H
	MOV BX, OFFSET X
	MOV AX, [BX]
	INC BX
	MOV CX,[BX]
	PUSH AX
	PUSH CX
	POP AX
	POP CX
	
	AX= 1A10H CX= 104F H

E. MOV BL,37H
MOV CX,7
MOV AX,0
MOV DL,1
L: ADD AX,CX
SHL DL,1
TEST BL,DL
LOOPNZ L
CX=CX-1
CX≠0 ZF=0
 	
AX= ~~[illegible]~~ 20H ; CX=...~~7~~ 00 H
DL=... ~~[illegible]~~ 00 H
~~20H~~
	F. MOV AX, 0004H
	MOV CL, 4H
	XOR BH, BH
	STC
	RCL BH, CL
	MUL BH
	
	BH= ~~[illegible]~~ 08 H
	AX= ~~20H~~
	2MD

Question #3 (8 points)

a) Suppose we have a microprocessor with 32bits address bus; what is the maximum memory capacity will be?

$2^{32} = 2^{2} \times 2^{30} = 4GB$

1

b) For the following instruction MOV [BX], 50H; what is the addressing mode used?

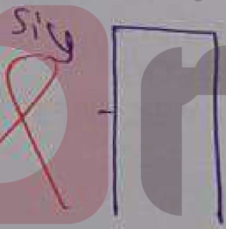
Base relative addressing mode

1

c) Write an assembler codes to clear 1KB of memory locations defined as byte.

X DB 1024, 2014..... 2^{10} = 1024 Byte

```
   MOV CX, 1024
Q: MOV Si, offset x
   mov BL, [Si]
   AND BL, 00H
   inc si
   dec cx
CMP CX, 1024
JL Q
HLT: Exit
```



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d) Write the equivalent code of the following instructions

- LOOP L

  Dec cx
  JNZ L1

  1

- SUB AL,BL

  XOR AL, BL

  0

- XOR AX,FFFFH

  NOT AX

  1